SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – SCENARIO BASED ASSIGNMENT

Course Code:	CSA17	**Course Title:**	Artificial Intelligence
CO Assessed:	CO2	**Assessment:**	Scenario Based Assignment
Weightage:	20% of CIA	**Duration / Marks:**	60 mins
CO2 Statement:	Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)		
Student Name:		**Reg. No.:**	
Section / Batch:		**Date:**	

Code of Conduct

I _____,_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Read each scenario carefully before answering (Off-Class Assessment)
- Answer all five scenario-based questions. Each question carries and is designed to be answered in about 12 minutes.
- Identify the key issues in each scenario, apply the relevant AI technique, and justify every decision with clear reasoning.
- Questions are drawn from Informed Search (Greedy, A*), Local Search and Optimization, Online Search, Constraint Satisfaction Problems, and Adversarial Search (Minimax, Alpha-Beta, evaluation functions).
- Marks will be awarded using the Scenario Based Assignment rubric: understanding of scenario, identification of key issues, application of concepts, decision making and clarity of response.
- Upload your work in GCR and Github.

Section / Topic	Focus	Question Nos.
1. Informed Search Strategies	Greedy best-first search, A* search, heuristic functions	**Q1**
2. Local Search and Online Search	Local search algorithms and optimization problems (hill climbing, simulated annealing, genetic algorithms), online search agent	**Q2, Q3**
3. Constraint Satisfaction Problem	Variables, domains, constraints, backtracking search, constraint propagation (forward checking)	**Q4**
4. Game Playing: Adversarial Search	Games, minimax algorithm, alpha-beta pruning, evaluation functions	**Q5**

Rubrics for Scenario Based Assignment

Criteria	Weightage	Level 4 – Exemplary (100% of marks)	Level 3 – Proficient (75%)	Level 2 – Developing (50%)	Level 1 – Beginning (25%)
Understanding of Scenario	**20%**	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	**20%**	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	**25%**	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	**20%**	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	**15%**	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria	Understanding of Scenario	Identification of Key Issues	Application of Concepts	Decision Making	Clarity of Response	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Question 1: Emergency Medicine Delivery Drones in a Flood Zone

Topic: Informed search strategies (Greedy best-first search, A, heuristic functions) | 12 minutes*

A government deploys AI-powered drones to deliver emergency medicines in a flood-affected region. The environment is highly dynamic: roads may suddenly become inaccessible, and weather conditions affect travel cost. The drone has access to:

- A heuristic estimate (straight-line distance) to the destination.
- Partial real-time updates about blocked paths.
- Variable movement costs depending on terrain and weather.

Due to urgency, the drone must quickly decide routes with minimum delay and risk.

Tasks:

(a) Explain how Greedy Best-First Search would behave in this scenario and discuss its strengths and weaknesses under uncertainty.

(b) Explain how A* would handle the same situation, including how it balances cost and heuristic.

(c) Analyze the impact of heuristic accuracy on both algorithms.

(d) Discuss how dynamic changes (blocked paths) affect search performance.

(e) Recommend the most suitable algorithm and justify your decision considering real-time constraints, optimality, and safety.

Question 2: Smart City Traffic Signal Optimization

Topic: Local search algorithms and optimization problems (hill climbing, simulated annealing, genetic algorithms) | 12 minutes

A smart city implements an AI system to optimize traffic signal timings across hundreds of intersections. The objective is to minimize:

- Total waiting time.
- Fuel consumption.
- Traffic congestion.

The system must continuously adjust signals based on traffic flow. However, the search space is extremely large, traffic patterns change dynamically, and the system may get stuck in locally optimal solutions.

Tasks:

(a) Formulate this as an optimization problem and explain why traditional search is inefficient.
(b) Explain how Hill Climbing would work in this scenario and identify its limitations.
(c) Discuss issues like local maxima, plateaus, and ridges with real-world interpretation.
(d) Explain how Simulated Annealing or Genetic Algorithms can overcome these limitations.
(e) Recommend the best approach and justify your choice based on scalability and adaptability.

Question 3: Autonomous Rover Exploring Mars

Topic: Online search agent | 12 minutes

An autonomous rover is deployed on Mars to explore unknown terrain. It must:

- Navigate safely to collect samples.
- Avoid hazards (craters, rocks).
- Operate with limited sensing capability.
- Make decisions without a complete map.

The rover updates its knowledge as it explores, making this a real-time decision-making problem.

Tasks:

(a) Explain why this problem requires an online search agent instead of offline search.

(b) Analyze the challenges of partial observability and dynamic environments.

(c) Describe how the rover builds and updates its internal model.

(d) Compare online search with uninformed and informed search strategies.

(e) Justify the most suitable approach considering uncertainty, adaptability, and safety.

Question 4: University Exam Timetable Generation

Topic: Constraint satisfaction problem (backtracking search, constraint propagation, forward checking) | 12 minutes

A university is designing an AI system to automatically generate exam timetables. The system must satisfy multiple constraints:

- No student should have overlapping exams.
- Limited number of exam halls.
- Certain subjects must be scheduled before others.
- Faculty availability constraints.
- Special accommodations for some students.

The complexity increases as the number of courses and students grows.

Tasks:

(a) Formulate the problem as a Constraint Satisfaction Problem (CSP): clearly define the variables, domains, and constraints with explanation.

(b) Explain how backtracking search works in this scenario.

(c) Discuss how constraint propagation (for example, forward checking) improves efficiency.

(d) Analyze real-world challenges and justify the best strategy for scalability.

Question 5: Strategic Game AI for Real-Time Decision Making

Topic: Game playing: adversarial search (minimax, alpha-beta pruning, evaluation functions) | 12 minutes

A gaming company is developing an AI opponent for a real-time strategy game. The AI must:

- Compete against human players.
- Make decisions within strict time limits.
- Evaluate complex game states with multiple possible moves.

The decision tree is extremely large, making computation expensive.

Tasks:

(a) Explain how the Minimax algorithm models this problem.
(b) Analyze the computational challenges of large game trees.
(c) Explain how Alpha-Beta pruning improves efficiency with detailed reasoning.
(d) Design an evaluation function and justify the factors considered.
(e) Recommend a strategy for balancing optimality and real-time performance.